Lab 3 — Forward/Reverse Controls: Plan

ELECT 435A (70156), Fall 2026. Written 2026-11-05 from the Canvas pages saved in this folder. The task-by-task checklist is in `lab3-tasks.org`.

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1. The most important thing: the real deadline is drop-off, not Nov 30

Canvas lists **Testing Prior to Power ON**, **Operation (video)** and **Procedure Component Install & Wiring (photos)** as due **Mon Nov 30**. But:

- **Controller Drop-off** is due **Tue Nov 24, 6 pm**, and the page says "*By now it is expected that your wiring is completed and operational.*"
- The listed drop-off windows are **Fri Nov 20 (1–9 pm)** and **Sat Nov 21 (8 am–2 pm)** in RR101.
- When the box comes back it has **4 faults installed**. You can't film a clean operation demo or take "as-built" photos after that.

So everything that needs a working, unmodified box has to be done **before you drop it off**. Your real target is a finished, tested, filmed and photographed box by **Thu Nov 19**, then drop-off Fri Nov 20 or Sat Nov 21. The Nov 30 dates only say when you have to upload.

2. Everything Lab 3 is graded on (672 pts total)

Listed in the order you will actually do them, not the order Canvas shows them.

#	Deliverable	Pts	Canvas due	When you actually do it	Type
1	Check In 1 (3 photos)	10	Mon Nov 16	Sun Nov 15	PDF

#	Deliverable	Pts	Canvas due	When you actually do it	Type
2	Testing Prior to Power ON (20-question quiz)	100	Mon Nov 30	Wed Nov 18, *before* power-on	Quiz
3	Operation (demo video, 10 tests)	100	Mon Nov 30	Thu Nov 19, before drop-off	Media
4	Procedure Compo-nent Install & Wiring (5 photos)	55	Mon Nov 30	Thu Nov 19, before drop-off	Image upload
5	Check In 2 (3 photos)	10	Mon Nov 23	Thu Nov 19 (take the photos before drop-off)	PDF
6	Controller Drop-off (text: your pick-up time)	10	Tue Nov 24	The day you drop off	Text
7	Selloff (instruc-tor grades your wiring)	100	Mon Dec 7	Graded while the instruc-tor has the box. Your part is the self-inspection before drop-off	—

#	Deliverable	Pts	Canvas due	When you actually do it	Type
8	Check In 4 (3 photos)	10	Sun Dec 6	After troubleshooting	PDF
9	Timed Troubleshooting (4-hr quiz + video)	2	Mon Dec 7	One 4-hour block after pick-up	Quiz + video
10	Forward/Reverse Lab Report	**275**	Sat Dec 12	Write it in pieces as you go, finish ~Dec 10	PDF

The report is 41% of the lab. The **Project Conclusion alone is 50 pts**, more than the operation video's wiring score.

3. Phases

Phase 0 — Orientation (Nov 5–8)

1. Watch, in this order:
   - `ELECT-435A-Lab3-Circuit [amJ3XY5agPo].mkv` (20:42): how the circuit works. Take notes on the ladder as you go.
   - `ELECT-435A-Lab3-Wiring [D7uTXLA_n8g].mkv` (6:49): installing and wiring.
   - Hold off on `ELECT 435A LAB3 OPERATION DEMO [AAsnMvPTKYE].mkv` (2:42) until you're ready to film. It's the model for your own video.
2. Study `LAB3-F20b(Ladder Diagram)-1.pdf` with `Additional Specifications…pdf`, `LAB3-COMPONENTS-1.PNG`, `LAB3-FRONT PANEL-1.PNG`, and `Proper Grounding of the Motor Control Transformer-2.pdf`.
3. Write your own wire list (a draft is in §8 of this file) and mark which wires cross the "door" (exactly 9).
4. Start the report file now: cover sheet and all section headers (see §7).

Phase 1 — Diagrams (Nov 6–10): 40 report pts you can earn before touching a screwdriver

- **Constructor diagram** of the ladder (20 pts). There are videos in the Introduction module: *Printing your Line Ladder Diagram with Constructor* and *Getting Line Ladder Diagrams into your Lab Reports*.
- **Hand-drawn diagram** (20 pts). Use a straightedge, label everything, and draw it neatly. "Crooked lines" and "not labeled" are the rubric's markdown reasons.

Phase 2 — Install components (Nov 7–10)

Follow the component layout from left to right: transformer, CB1, TBG1–3 (green), TB1–3, TB4–6 (24V-N), TB7–9 (24V-H), CB2, 3PHIN (CR socket), FR (FW) and RV contactors with OLF/OLR, 3PH POWER terminals TB-T1..T3. Put **end stops between every component** (not between terminal blocks). Door layout from the front: STOP1, FWD, STOP2, REV, selector (unused), FLS, RLS, FWDON, REVON, AC IN.

Phase 3 — Wire it (Nov 9–17)

Suggested order, one rung at a time, checking each wire off on the ladder: 1. 120 V input: cord → CB1, TB1, TBG1 → transformer primary (black/white/green). 2. Transformer secondary + grounding: X1 → CB2 → TB7 (24 bus); X4 → TB4 (X4N bus); green from TBG1 **directly to X4 on the transformer** (see grounding PDF). 3. Rung 5–6 (FWD), rungs 7–8 (REV), rungs 9–10 (pilot lights). 4. Power circuit: 3PHIN socket → FW/RV line side (RV swaps L1 and L3) → OLF/OLR → TB-T1..T3.

Check In 1 photos on Sun Nov 15, whatever state the box is in. Points are for following the instructions and showing progress, not for being finished.

Phase 4 — Self-inspection, then the test quiz (Nov 17–18)

1. Do the **selloff self-inspection** in §6 first. The selloff is 100 pts and gets graded while the instructor has your box.
2. Make a **paper testing worksheet** (20 rows: question, meter leads, expected, actual). **The report gives credit only for the original sheet you filled in, not a retyped one.**

3. Take the *Testing Prior to Power ON* quiz with the power **off**, writing each reading on the sheet. Repeat until you get 100%. Write down anything you fix ("Items you fixed" is a required report item).
4. **Screenshot / print the quiz** for the report.

Phase 5 — Power on, film, photograph (Nov 19)

1. Power up only after scoring 100% on the test quiz.
2. Print `LAB 3 OPERATIONAL DEMONSTRATION.pdf`. Rehearse every test, then watch the instructor's demo video.
3. Film in **horizontal** format, close up, showing the buttons, lights and contactors. **Start with 5 s of your phone showing the date and time.** Film all **10** rubric tests (see §5). The printed sheet lists only 8.
4. Take the 5 Procedure photos (§4) and the Check In 2 photos. Upload them now or any time before the due date.

Phase 6 — Drop-off (Fri Nov 20 or Sat Nov 21; hard limit Tue Nov 24, 6 pm)

- Email Tedde Titus (ttitus@lbcc.edu) **this week** to set up drop-off **and** a pick-up appointment.
- **Do not take anything apart.** Submit the Drop-off text entry with your pick-up time.

Phase 7 — Timed troubleshooting (after pick-up; due Mon Dec 7)

1. **Before** starting: re-study the ladder, have your wire list and meter ready, and block off 4 uninterrupted hours.
2. Starting the quiz starts a 245-minute clock. Set your own timer for just under 4 hours.
3. Keep a **fault log as you go** in the report's table format: *Error # / Description / Problem caused / How you found it / What you did to fix it*. Be **very** detailed (30 pts in the report).
4. **One fault is a transformer problem.** The report asks specifically how you found it with a meter (20 pts). Write down your exact meter steps when you find it.
5. Answer the 2 quiz questions, film an operation video of the repaired box (same tests), and upload it to "*Lab 3 – Timed Troubleshooting Video Upload*". That assignment is **not** in the module list, so look for it under Assignments.

6. Check In 4 photos (due Sun Dec 6).

Phase 8 — Finish the report (due Sat Dec 12; aim for Thu Dec 10)

Needs your selloff results from the instructor. Ask for them if they haven't appeared by about Dec 8.

4. Photo checklists

Check In 1 / 2 / 4 (same spec each time, PDF: title page with name, class and lab number + 3 cropped photos): 1. Front view of controls 2. Rear of control panel 3. Top-down view of wiring area. Graded on: all 3 views (3), following the wiring instructions (3), components installed/wired correctly (3), progress since last week (1).

Procedure Component Install & Wiring (in this order): 1. Front panel: all buttons and lights in one shot 2. Rear of front panel, wiring and cable trays **open** 3. FWD/REV starter area: everything right of CB2, between the two horizontal trays, **covers off** 4. Transformer area: transformer to left of CB2, **transformer grounding clearly visible** 5. Overview: top-down, cable trays **installed**

5. Operation video: 10 tests and what should happen

Graded 8 pts each + 20 for wiring/neatness. The printed sheet has 8 tests, but the rubric grades 10. Film all 10.

#	Test	Steps	Expected
1	FWD & STOP1	FWD, STOP1	FW pulls in + FWDON on and stays on; STOP1 drops it
2	REV & STOP2	REV, STOP2	RV pulls in + REVON on; STOP2 drops it
3	OLF	FWD, press OL test on OLF	FW drops, FWDON off. Reset OLF afterward

#	Test	Steps	Expected
4	OLR	REV, press OL test on OLR	RV drops, REVON off. Reset OLR
5	FWD limit	FWD, RLS, FLS	RLS does nothing; FLS drops FW
6	REV limit	REV, FLS, RLS	FLS does nothing; RLS drops RV
7	FWD interlock	FWD, REV, STOP2	REV does nothing while FW is on (FW 31-32 open); STOP2 stops
8	REV interlock	REV, FWD, STOP1	FWD does nothing while RV is on; STOP1 stops
9	FWD & STOP2	FWD, STOP2	Runs, STOP2 drops it
10	REV & STOP1	REV, STOP1	Runs, STOP1 drops it

6. Selloff self-inspection (100 pts, 10 each)

- ☐ **Neat wiring**, up to industry standard
- ☐ **No exposed copper** at any terminal (tug and look at every one)
- ☐ **No loose wires**: re-torque every screw. Loose wires found at inspection become *extra* faults in your troubleshooting.
- ☐ **Wire routes**: all front-panel control wires in the **right** cable tray, none in the left tray to the panel, **no solo neutrals** (hot and neutral run together)
- ☐ **3-phase wired** between CR socket and terminal blocks (all of it, not half)
- ☐ **Exactly 9 wires** through the door channel
- ☐ **Components in the proper location** per the layout picture
- ☐ **Cable trays**: wires routed properly, not overstuffed, no bypassing, not too tight
- ☐ **All cable tray covers on**
- ☐ **Proper colors**: 24 V hot red, 24 V neutral white, 120 V hot black, 120 V neutral white, 3-phase black, ground green, all 16 AWG stranded

Also from the general spec: wires in the top and out the bottom; max 2 wires per terminal; **1 wire per circuit-breaker terminal**; no wire nuts; nothing on unused terminals; the transformer primary feeds nothing else; end stops between components.

7. Report outline (275 pts), in the required order with required headers

Section	Contents	Pts
Cover sheet	Standard	5
WIRING SECTION	Hand-written diagram (20); Constructor diagram (20); Summary ≥2 ¶ (10); Conclusion ≥2 ¶ (20)	70
Testing Worksheets	Scan/photo of the *original* paper sheet, expected vs actual	20
TESTING SECTION	Testing-process description + quiz printout; Items you fixed; Summary ≥2 ¶ (10); Conclusion ≥2 ¶ (20)	30
TROUBLESHOOTING SECTION	Errors Found Table (10 format + 20 content); Transformer problem Q&A (20); Summary ≥2 ¶ (10); Conclusion ≥2 ¶ (20)	80
SELL-OFF SECTION	Points lost and why, with your final score, even if no points were lost (20); **Project Conclusion**: substantial, well beyond 2 ¶, concrete learnings backed by examples (50)	70

Summary = what you did. Conclusion = what you learned, with evidence. Label every item so the instructor never has to guess which one they're reading.

Write each section's draft **right after** that phase ends, while it's fresh.

8. Draft wire list (derived from LAB3-F20b; verify against your sheet)

Feed-through TBs: top and bottom screws are the same point. TB7–9 are bussed (24 V hot) and TB4–6 are bussed (X4N), as in Lab 2.

Wire	Color	Connects	Crosses door?
H	black	Cord hot → CB1-1; CB1-2 → transformer H1	
N	white	Cord neutral → TB1; TB1 → transformer (N1)	
G	green	Cord ground → TBG1; TBG1 → **X4 at the transformer**	
X1	red	Transformer X1 → CB2-1	
24	red	CB2-2 → TB7	
X4	white	Transformer X4 → TB4	
1	red	TB7 → STOP1-1	**yes**
2	red	STOP1-2 → STOP2-1	door only
3	red	STOP2-2 → FWD-3 → REV-3 (door); → FW 7L4 → RV 7L4 (panel)	**yes**
4	red	FWD-4 → FLS-21 (door); → FW 14/22	**yes**
5	red	FLS-22 → RV 31	**yes**
6	red	RV 32 → FW A1	

Wire	Color	Connects	Crosses door?
7	red*	FW A2 → OLF 95	
8	white	OLF 96 + OLR 96 → TB5	
9	red	REV-4 → RLS-21 (door); → RV 14/22	**yes**
10	red	RLS-22 → FW 31	**yes**
11	red	FW 32 → RV A1	
12	red*	RV A2 → OLR 95	
13	red	TB8 → FW 23	
14	red	FW 24 → FWDON X1	**yes**
15	white	FWDON X2 → REVON X2 (door); → TB6	**yes**
16	red	TB9 → RV 23	
17	red	RV 24 → REVON X1	**yes**

Door-crossing count: 1, 3, 4, 5, 9, 10, 14, 15, 17 = **9** ✓. This only works if you make the jumpers on the door (STOP1→STOP2, STOP2→FWD→REV, FWDON X2→REVON X2), so each signal crosses once.

*Wires 7 and 12 sit upstream of the overload contact, so they're *not* grounded conductors. That's the same question as wire 4 in Lab 2. Use whatever color your instructor accepts for this.

Power (all black): 3PHIN socket L1 (pin 24), L2 (21), L3 (14) → FW 1L1/3L2/5L3 = L1/L2/L3 and RV 1L1/3L2/5L3 = **L3**/L2/**L1** (the L1–L3 swap reverses rotation). OLF and OLR 2T1/4T2/6T3 tie together → TB-T1/T2/T3. The contactor's 13 = 7L4 and 14 = 14/22, as noted on the ladder.

9. Clarification - ask the instructor

1. **Testing/Operation/Procedure are due Nov 30, but the box is dropped off by Nov 24.** Confirm they must be done before drop-off. I'm assuming they do.

2. **Pick-up is “Thursday or Friday”**, which is probably **Nov 26–27, Thanksgiving**. Confirm the actual pick-up days.
3. **Drop-off**: the page says “by Tuesday 6 pm” but only lists Fri/Sat windows. Is Tuesday an option?
4. **Check In 3 doesn’t exist** in the module (1, 2, 4 only). Is that intentional?
5. **Timed Troubleshooting Video Upload** assignment is referenced but isn’t in the module list.
6. **Operation video**: the page says 9 scenes, the printed sheet has 8 tests, and the rubric grades 10. Film all 10.
7. **Testing worksheet**: there’s no worksheet file. Is a self-made paper sheet OK?
8. Two slightly different report specs exist (the “REPORT SPECIFICATIONS-2” page vs. the report assignment). **Follow the assignment page**, since it has the rubric.

10. Other deadlines in the same window

Test #2 (Start/Stop/Jog) and Ch. 8 HW **Mon Nov 16** (same day as Check In 1) · Ch. 7 HW Mon Nov 23 · Ch. 12 HW Sun Dec 6 · **Final Exam Mon Dec 14**.